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Review article

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Sleuthing subjectivity: a review of covert measures of consciousness

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Abstract

Consciousness is private. Although conscious beings directly access their own conscious experiences, the consciousness of others must be inferred through overt report: observable behaviours — such as overt facial expressions, vocalizations and body gestures — that suggest the level, state and content of consciousness. However, overt report is limited because it can be erroneous (for example, resulting from wilful deception or being subject to recall error), absent (for example, during sleep and paralysis) or conflict with research goals (for example, in no-report paradigms and resting-state studies). These limitations encourage the search for covert measures of consciousness: physiological signals that disclose consciousness without relying on overt behaviour. This Review highlights emerging covert measures of consciousness in humans, including eye, skin, respiratory and heart signals. We also address the challenge of distinguishing physiological signals linked to conscious versus unconscious neural processing. Finally, we consider the ethical implications of infringing on the innate privacy of consciousness.

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Introduction

Consciousness is the first-person, subjective experience of the world and body, including 'what it is like' to see, dream and feel an emotion[1]. In essence, there is 'something it is like' to be conscious[2]. Any specific instance of this experience is directly accessible only to the conscious being themself — the subject of those conscious experiences — while being inaccessible to external observation. Consequently, detecting consciousness in others is challenging. Although we cannot directly experience the consciousness of others, we infer their subjective states through conscious or overt report: observable and often voluntary behaviours — such as overt facial expressions, vocalizations and body gestures — that suggest the presence and details of conscious experience (Fig. 1a). In short, overt report enables the conscious being to divulge their subjective states and the observer to infer these conscious experiences.

Overt report appears to be a reliable and definitive method — perhaps the only approach — to access the consciousness of others. In practice, however, overt report is a limited solution to overcome the subjective nature of consciousness. The limitations of overt report motivate the search for covert measures of consciousness: physiological signals that can be used to infer conscious experience.

Covert access to consciousness carries scientific, clinical, ethical and societal implications. In science, covert measures of consciousness can be implemented when examining the biological mechanisms of consciousness, including enabling the design of experimental tasks that limit the influence of report-dependent processes[3]. Clinically, covert measures of consciousness may aid the diagnosis and treatment of people with an uncertain level or content of consciousness[4,5], and their application in the operating room could be valuable for detecting intraoperative awareness[6]. Additionally, combining covert measures of consciousness with overt report can provide a more reliable assessment of patient symptoms (for example, pain severity). Beyond human applications, species-specific covert measures may help to infer non-human animal consciousness[7]. However, beneficial applications must be weighed against the ethical dilemmas introduced by the risk that covert access to consciousness undermines the inherent privacy of subjective experience.

This Review has several objectives. First, we highlight the limitations of overt report that motivate the need to identify covert measures of consciousness. Second, we propose a standardized definition for covert measures of consciousness. Third, we review existing literature that supports various categories of physiological signal — such as eye metrics, dermal signals and cardiovascular signals — as potential covert measures of consciousness. Last, we explore the major challenges for identifying covert measures and emerging ethical dilemmas related to encroaching on the privacy of consciousness.

The limitations of overt report

Overt report is associated with several major limitations. First, overt report can be erroneous or deviate from the veridical presence or content of conscious experience (Fig. 1b). Overt report can misguide inferences about consciousness owing to (1) incomplete overt reporting (for example, insufficient introspection for reporting on highly detailed or abnormal conscious experiences[8–10]), (2) voluntary inaccurate overt reporting (for example, censoring or fabricating conscious experiences to gain an advantage or align with perceived social norms[11,12]), (3) involuntary inaccurate overt reporting (for example, distortion and neglect during the encoding, storage and recall of conscious experiences, such as in faulty eye-witness testimony, confabulation or experimental-based manipulations[13–15]), and (4) the appearance of overt report without consciousness (for example, speech and text generated by large language models, gestures in unresponsive wakefulness syndrome, or pathological crying and laughing in patients with brain injury who are suspected of having a low conscious level[16,17]).

Second, even assuming that overt report is veridical, it is not always available to observe (Fig. 1c). For instance, overt report is absent during sleep and altered states of consciousness (for example, stupor, seizure and general anaesthesia), and in disorders that cause paralysis (for example, locked-in syndrome). The estimated high misdiagnosis rate in disorders of consciousness is partly attributed to the limited or absent ability of individuals in this patient group to provide overt report[18]. Even encounters between healthy, awake humans who do not speak the same language can hinder the disclosure and interpretation of subjective states. Similarly, the conscious experiences of preverbal children can be challenging to determine[19]. Moreover, the inherent language barrier between humans and non-human animals — except for a few possible exceptions[20] — makes interpreting non-human animal behaviours as definitive markers of consciousness difficult, which contributes to ongoing debates on the ethical status of non-human animals[7].

Third, even assuming that overt report is veridical and always available to observe, it can conflict with research goals (Fig. 1d). When examining the biological mechanisms of consciousness, overt report can confound physiological recordings with post-perceptual processes (for example, decision-making, working memory and locomotion). Discriminating between consciousness-dependent and report-dependent physiological processes is challenging and hotly debated[21,22]. An approach to manage this issue is through no-report paradigms (for examples, see refs. 3,23), which minimize or eliminate task-based overt report with the goal of reducing the engagement of post-perceptual processing. However, it is debated to what extent overt report introduces confounding signals and whether no-report paradigms are optimal to resolve this potential challenge in isolating the neural mechanisms of consciousness[24,25]. Furthermore, in resting-state experiments, the goal is to study task-independent or spontaneous physiological activity, so overt report undermines the idle physiology of interest. Yet, without overt report, the subjective states that could help to explain task and rest-dependent physiology are neglected[26].

Covert measures of consciousness

There is no standard definition for covert measures of consciousness. Here, we propose that such measures should be (1) involuntary, (2) accessible and (3) informative of consciousness (Fig. 2a). By meeting all three criteria, covert measures are less affected by the limitations of overt report while still being useful for inferring details of consciousness.

The first criterion for a covert measure of consciousness is that it is involuntary. This criterion is critical, because the volitional nature of overt report gives rise to several limitations, including that conscious reporting can be voluntarily inaccurate (that is, it can result from deception). By contrast, involuntary physiological signals, such as pupil size and skin conductance, are not similarly susceptible to wilful manipulation. Meanwhile, some physiology has both voluntary and involuntary modes (for example, blinking, eye movements and respiration). It is important to distinguish between measures that are fully involuntary versus those that can be directly influenced by voluntary actions to help judge their reliability as covert measures of consciousness. In addition, assessing the dynamic between voluntary and involuntary influences on covert measures is relevant for dissociating the neural

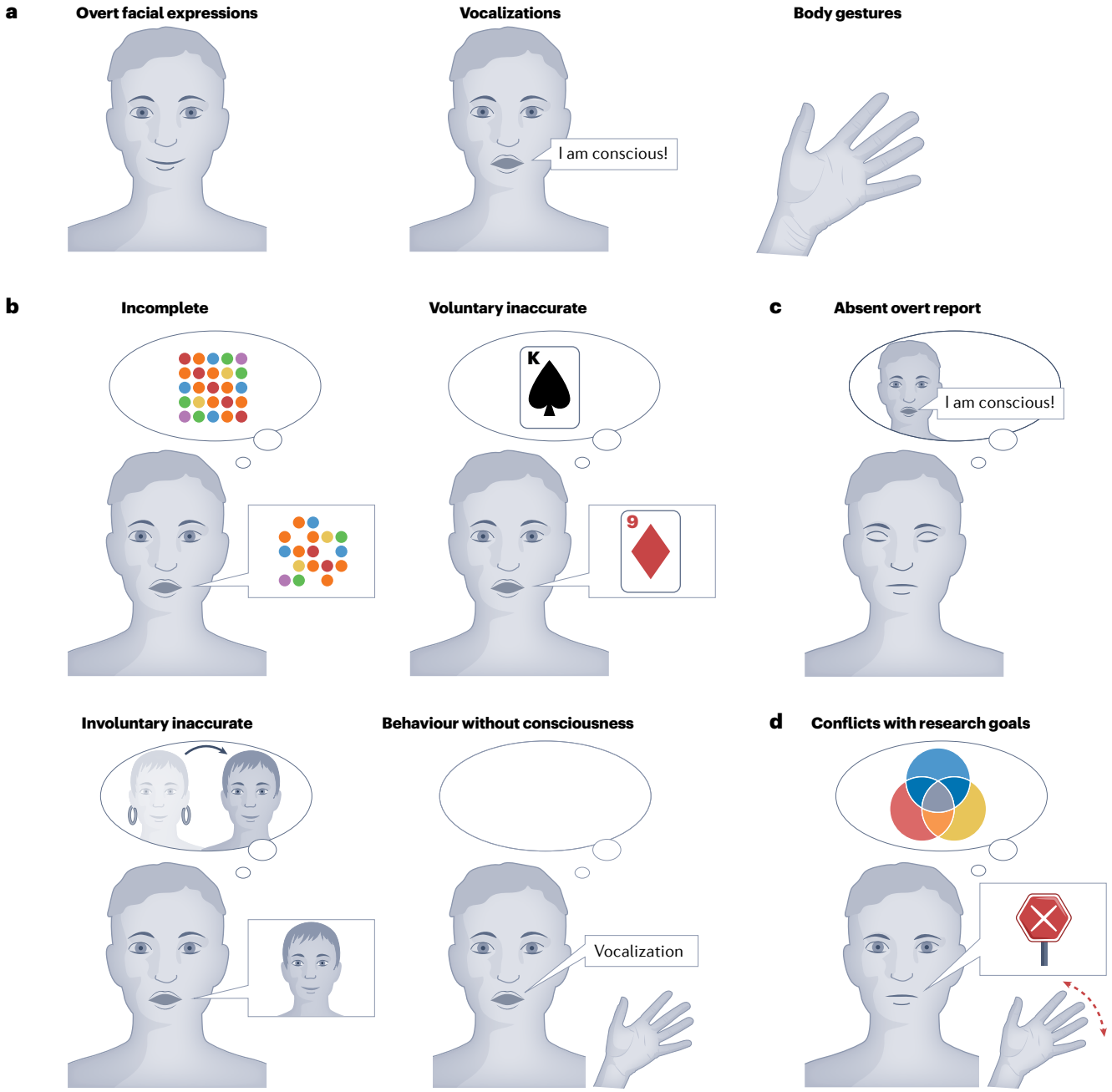


Fig. 1 | Methods of overt report and limitations. a, Overt reporting methods include overt facial expressions (for example, smiling), vocalizations (for example, speech) and body gestures, particularly voluntary movements (for example, hand waving). **b**–**d**, There are three primary limitations of overt report: a report can be erroneous (**b**) absent (**c**) or in conflict with research goals (**d**). Overt report can be erroneous relative to the veridical status of consciousness. Common types of erroneous overt report include incomplete overt report, voluntary inaccurate overt report, involuntary inaccurate overt report and behaviour without consciousness (**b**). Incomplete overt reports are cases of insufficient introspection (for example, a rich conscious content that is reported in only partial detail). Voluntary inaccurate overt report is when the reporter intentionally misrepresents their conscious experiences (for example, fallaciously reporting a card that the reporter is imagining in a game of deception). Involuntary inaccurate overt report is when, unbeknown to the reporter, a conscious experience is recalled inaccurately owing to errors in encoding, storage and/or recall processes (for example, initially perceiving a person with earrings but later recalling them without earrings). Behaviour without consciousness comprises cases when overt actions with the appearance of overt report occur in the absence of consciousness (for example, the movements and vocalizations of patients with unresponsive wakefulness syndrome). Even assuming that overt report is always veridical, the regular absence of overt report (for example, during sleep) limits its reliability for deducing consciousness (**c**). Even assuming that overt report is always veridical and present, overt report can conflict with research goals (for example, in no-report paradigms and resting-state experiments) that preclude engaging in behaviours that may confound measures of interest (for example, overt reporting introduces post-perceptual processes that can challenge interpretion of the neural mechanisms of consciousness) (**d**).

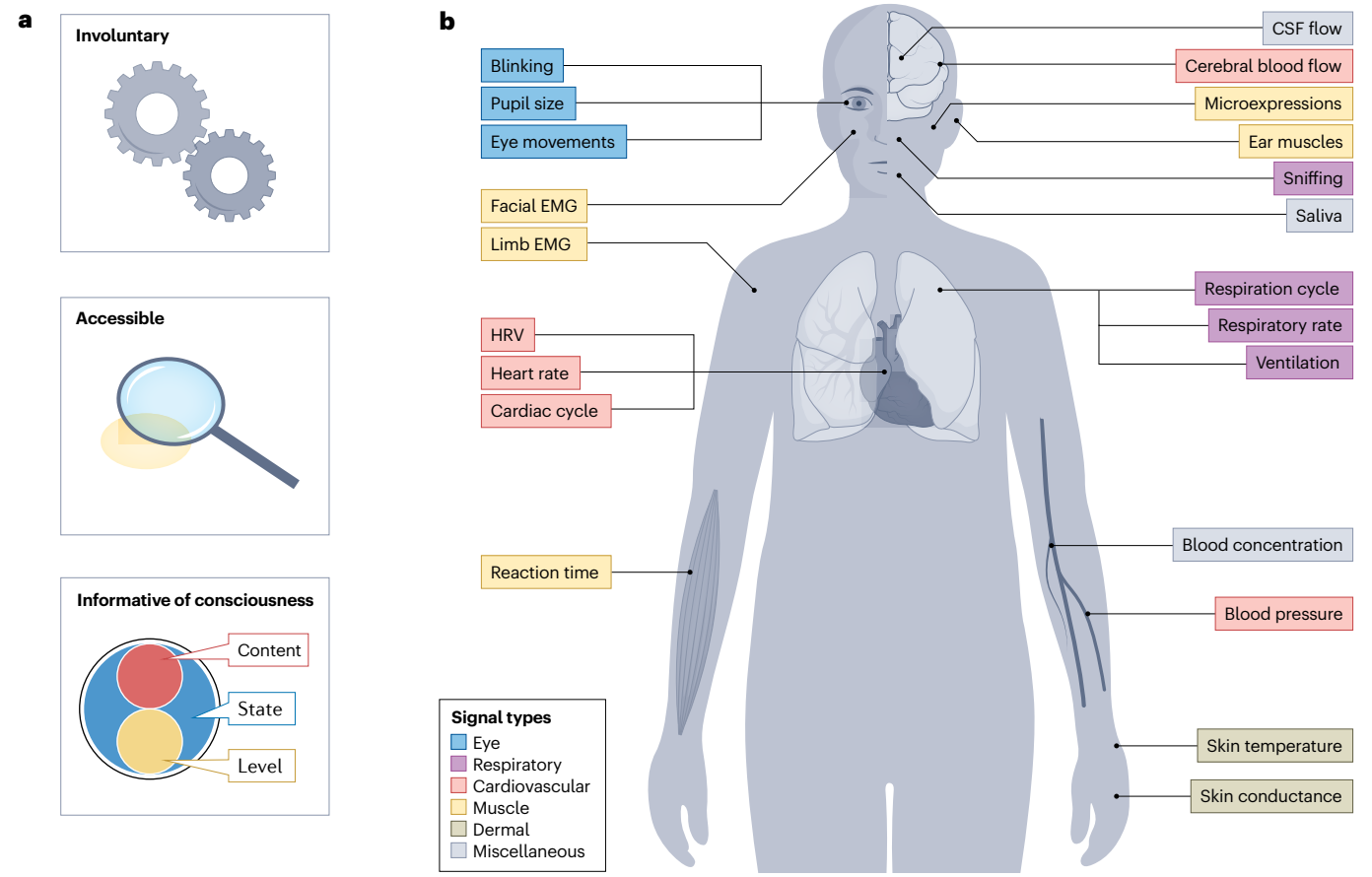


Fig. 2 | Covert measures of consciousness. a, We argue that a covert measure of consciousness should meet three main criteria: it should be involuntary, accessible and informative of consciousness. To meet the involuntary criterion, depicted here with gear icons to represent the absence of volitional control or automaticity, physiological signals should be modulated without wilful control. Note that, however, this criterion does not exclude physiology with both voluntary and involuntary modes of regulation (for example, blinking and respiration). To meet the accessible criterion, physiological signals should be easily recordable and interpretable, thus excluding bodily measures that require invasive procedures (for example, intracranial neural recordings). To meet the informative of consciousness criterion, physiological signals should predict at least one of three major dimensions of consciousness: conscious level, conscious content and conscious state. **b**, A schematic showing highlighted covert measures of consciousness grouped by signal type, namely (1) eye signals (for example, pupil size, blinking and eye movements), (2) muscle signals (for example, microexpressions, facial and limb electromyography (EMG), ear muscle signals and reaction time), (3) dermal signals (for example, skin temperature and skin conductance), (4) respiratory signals (for example, sniffing, respiratory cycle and rate, and ventilation), (5) cardiovascular signals (for example, cardiac cycle, heart rate, heart rate variability (HRV), blood pressure and cerebral blood flow) and miscellaneous signals (for example, cerebrospinal fluid flow (CSF), saliva, and blood concentration of free fatty acids and adrenaline). Covert measures of consciousness can be used independently or combined with overt reporting to guide more accurate inferences on human consciousness. The highlighted covert measures of consciousness may also be applicable in non-human animals, although species-specific physiological signals must be considered.

mechanisms of consciousness from post-perceptual processes that are often related to voluntary actions (for example, deciding which responses to make during an experimental task). Likewise, the involuntary criterion excludes methods that rely on voluntary engagement in covert cognitive or perceptual tasks (for example, solving arithmetic or imaging a scene to infer responses to questions from ensuing physiological signals[27,28]). Although these approaches are covert relative to overt speech and gestures, they still depend on wilful participation and faithful introspection, and thus are susceptible to limitations that are similar to those of overt reporting.

The second criterion is that a covert measure of consciousness is accessible. The primary advantage of overt report is ease of observation. However, the limited capacity of healthy human senses restricts the repertoire of behaviour and physiological signals that qualify as overt. By contrast, covert measures of consciousness extend beyond what is readily observable, provided the physiological signal can be reliably measured with recording instruments (for example, eye tracking and heart rate monitors). We also define accessibility by how easily candidate signals can be collected and analysed. This Review omits signals requiring invasive procedures or highly specialized equipment that precludes widespread implementation. Therefore, this criterion excludes many neural signals as potential covert measures of consciousness (see the 'Neurophysiology' section below). Still, the accessibility of physiological signals is dynamic according to the

subject group and advancements in recording technology. In short, an inaccessible physiological signal today may become accessible in the future.

The third criterion is that a covert measure of consciousness is informative of consciousness. Importantly, the precise mechanism by which a physiological signal reflects conscious experience is not essential for designating it as a covert measure. For instance, there is debate over whether physiological signals are lagging indicators or causal factors of conscious experience (for example, eye movements or blinks indicating versus causing perceptual switches to a bistable image[29,30]; baseline pupil size predicting perceived stimulus brightness, related to the amount of photons that reach the retina[31]). Likewise, it is unresolved whether physiological signals are directly modulated by the neural mechanisms of consciousness or by secondary processes (see the 'Perspectives' section for additional discussion). Through either direct or indirect relationships, a covert measure can still be informative of consciousness. For example, a marker of arousal — a physiological state that is distinct from, yet an enabling factor of consciousness — can help one to infer the likelihood that a subject is conscious.

Covert measures can inform three primary domains of consciousness (Fig. 2a): (1) conscious level, (2) conscious content and (3) conscious state[32,33].

Conscious level measures physiological states that predict the likelihood of being or becoming conscious, which are closely linked with arousal, alertness and vigilance (see ref. 34 for a discussion on the limits of using levels of consciousness as a theoretical construct)[35]. For instance, conscious level differs in healthy wakefulness versus drowsiness, sleep, general anaesthesia and the disorders of consciousness.

Conscious content is the specific subject of conscious experience, such as seeing a familiar face, hearing wind chimes or feeling happy. The content of consciousness also includes sensations that underlie agency (for example, intention and control) and embodiment (for example, possession of a body) that are linked with self-consciousness: the subjective experience of being a self[36]. In other contexts, particularly the philosophy of mind, conscious content is called qualia, phenomenology or phenomenal experience[37].

Conscious state represents a unique combination of conscious level and content. If conscious level and content chart a two-dimensional plane, conscious state defines a position within this space. For instance, drug-induced psychedelic states often combine high conscious level with synaesthetic conscious content. Meanwhile, other normal and altered states of consciousness (for example, fatigue, daydreaming and hypnosis) occupy unique combinations of conscious level and content.

In this Review, we organize findings on the informativeness of potential covert measures of consciousness in two sections: (1) a combined section on conscious level and state and (2) a conscious content section. The conscious level and state section includes findings on precursor or enabling factors of consciousness (for example, arousal and attention) that help to facilitate conscious level or content but may not be necessary or sufficient for consciousness[21]. In the conscious content section, we highlight research on physiological signals that indicate conscious access (for example, indication of a perceived versus not perceived sensory event), independent of its phenomenal experience or content. We categorize research on covert measures that are used to infer conscious access with those that directly indicate conscious content because, in many experimental contexts, there is knowledge of environmental or task parameters (for example, stimulus timing and type) that can be leveraged with information about conscious access to infer the content of consciousness. For instance, the conscious content of a face is predicted by the indication of a conscious perception at the time that a known face stimulus appeared[3] (Fig. 4a).

Finally, this Review predominantly highlights positive associations between physiological signals and conscious level, state and content to help guide the field towards the most promising covert measures of consciousness. Future research and literature reviews should explore the limits of candidate covert measures of consciousness, including reproducibility across environments and participant groups.

Categories of covert measures of consciousness

We highlight several major categories of physiological signal that may serve as covert measures of consciousness (Fig. 2b), namely (1) neurophysiology, (2) pupil size, (3) blinking, (4) eye movements, (5) muscle signals, (6) dermal signals, (7) cardiovascular signals and (8) respiratory signals. Other physiological signals linked with consciousness, including blood and salivary signals, are discussed in Box 1.

Neurophysiology

Neurophysiology provides a rich source of information about human consciousness. Conscious level and state, including sleep, general anaesthesia and disorders of consciousness, are inferred from multiple measures of neurophysiology (for example, neuronal spiking activity[38], functional connectivity[39,40], measures of brain complexity[41] and brain metabolism[42–44]). Neural decoding predicts and reconstructs conscious content, such as visual conscious perception during imagery and dreams[45–48], covert movements (for example, imagined handwriting)[49], sensory stimulation (for example, sounds and images)[50–53], and illusory, bistable and hallucinatory conscious perception[54,55].

However, despite its value, neurophysiology has practical limitations. Relative to the acquisition of many other physiological signals, the recording of neurophysiological data can be time consuming, expensive, invasive, exclusionary of certain groups (for example, people with claustrophobia, incompatible implants and the inability to remain still) and restricted to specialized recording environments. Moreover, the data that are generated can be complex to analyse. In addition, using neurophysiology as a covert measure of consciousness

Box 1 | Additional categories of covert measures of consciousness

The physiological signals highlighted in the main text represent a partial list of potential covert measures of consciousness. Other candidate physiological dynamics that might be linked to consciousness include eye weeping[235], cerebrospinal fluid flow (which may show differences between wakefulness and sleep[419]), the blood concentration of free fatty acids (which may be correlated with dream anxiety content[420]) and adrenaline (which shows a decrease during general anaesthesia[272]). In addition, spontaneous saliva management varies between conscious levels in disorders of consciousness[239], and saliva flow can predict the conscious perception of taste (for example, imagining bitter foods)[421,422]. In fact, salivary pH has been used to establish communication with a locked-in patient, who responded 'yes' or 'no' by imagining tasting lemon or milk, respectively[423]. Additional covert measures of consciousness are currently under investigation.

is controversial because the precise make up of the consciousness brain network is debated and can confound research programmes studying the neural mechanisms of consciousness. For instance, relying on a specific brain region or neural activity pattern as a marker of consciousness may bias results, reinforcing the presupposition that this neural marker is linked to consciousness. These constraints highlight the need to explore non-neural physiology measures that are featured in this Review, which may offer more accessible and versatile covert measures of consciousness.

Pupil size

Pupil size regulates retinal illumination: the amount of light that accesses photosensitive cells in the back of the eye. Likewise, the pupillary light reflex involves changes in pupil size in response to light stimulation (for example, dilating in darkness). Beyond light responses, pupil size is linked to certain patterns of neural activity at the level of neuronal spiking and large-scale brain network activity[56–59]. This pupil–neurophysiology link explains why pupil size correlates with behaviour and cognition[60,61]. In addition, pupil size reflects conscious level, state and content. Pupil size is an ideal physiological signal because it can be recorded in diverse environments using relatively inexpensive camera equipment. However, a challenge for recording pupil size is the frequent occlusion of the pupil by the eyelid, such as during blinks and sleep. Additionally, head and eye movements can distort the pupil image, complicating accurate measurement[62].

Conscious level and state. Conscious level is closely linked to arousal — an enabling factor of consciousness. For example, arousal level helps to determine states of drowsiness, sleep, general anaesthesia and disorders of consciousness[63]. Transient shifts in arousal are also hypothesized to coordinate attention and sensory networks that are relevant for conscious perception[64,65]. Pupil size is linked to neuromodulatory regions that regulate chronic and transient changes in arousal, and thus is informative of conscious level[66,67]. Additionally, the link between pupil size and salience and sensory brain networks further binds pupillary dynamics with conscious level, state and content (see below)[3,68]. For example, pupil size reflects levels of wakefulness and fatigue[69,70]. Yet more, both spontaneous and evoked, pupil size dynamics (for example, responsiveness to light stimulation) are diagnostically and prognostically important in numerous neurological disorders, particularly in disorders of consciousness[71,72].

Moreover, pupil size is sensitive to altered states of consciousness, including hallucinogenic[73] and mind-wandering conscious states[74]. Pupil size also predicts perceptual sensitivity — a proxy of conscious level that measures the likelihood that current and future sensory stimuli will be consciously perceived[75,76]. In support of this, pupillary responses to auditory stimuli discriminate between minimally conscious state and unresponsive wakefulness syndrome[77].

Conscious content. Conscious content can be inferred from pupil size through the pupillary light reflex. For instance, one study used binocular rivalry with images of low and high luminance[78]. Pupil size tracked the consciously perceived image and perceptual switches between images — dilating when perceiving the darker image and constricting when perceiving the lighter image. Similarly, pupillary responses to a flashing light were greater in the dominant eye (that is, the eye receiving input for the consciously perceived image) than the suppressed eye during binocular rivalry[79]. Moreover, covert attention to light or dark stimuli, even to images placed on light or dark backgrounds, results in corresponding pupil size changes[80–82].

Pupil size also tracks the occurrence of a perceived sensory event, providing clues about conscious perception when combined with knowledge of the sensory environment (for example, the timing and type of images shown to participants[3,83,84]; Fig. 4a).

Furthermore, pupil size is sensitive to perceived luminance or brightness, independent of physical light intensity. For example, the pupil constricts in response to the illusion of brightness and dilates for perceived darkness[85,86]. This effect relies on conscious perception[87]. In fact, pupil size is sensitive to the imagery of dark or light objects and scenes[88–90]. Even words with the connotation of brightness or darkness (for example, 'sun' and 'night') induce predictable pupillary responses — more constricted for words conveying brightness[90]. This effect may be present because participants are imagining the word content. However, people with limited or no imagery ability do not show these pupil size changes related to luminance imagery[91]. This suggests that the conscious content of luminance is essential for pupil size change in the absence of physical light stimulation.

Pupil size is also responsive to other features of conscious content, including stimulus size (real and illusory)[92,93], colour[94], motion (real and illusory)[95–97], shape (for example, dilating more for the illusory perception of a concave versus convex image)[98], distance (real, illusory and imagined)[99–101], numerosity (real and illusory)[102], perceptual switches[23,103–108], appearances and disappearances (real and illusory)[109], volitional decision information (for example, 'yes' versus 'no' or 'go' versus 'no-go' responses) and timing[60,110,111], locomotion (real and imagined)[112,113], perceived tactile stimulation (for example, pain intensity)[114–116], recognition[117] and emotion[118–122]. Pupillary responsiveness to illusory and imagined conscious content supports its sensitivity to subjective experience, independent of objective stimulus characteristics or even in the absence of sensory input.

Additionally, pupil size oscillations can predict covert, conscious attention to frequency-tagged images (that is, flickering stimuli at predetermined rates of change)[84]. This approach has been extended as a method to covertly communicate with healthy participants (for example, responding 'yes' or 'no' by attending to one of two images with distinct frequency characteristics)[123]. Likewise, pupil size fluctuations have been used to indicate which ear a participant was covertly attending to during dichotic listening[124] and to classify which song a participant was imaging[125].

Blinking

The brief closure of the eyelid or blinking (up to ~50 times a minute[126]) helps maintain eye moisture and guard the eye from damage (for example, reflexive eye closure in response to threat)[127]. Blinking is also influenced by cortical and subcortical activity, linking blinks with a wide range of behavioural, cognitive and perceptual processes[128–136]. One key function of central neural monitoring and regulation of spontaneous blinking is to achieve conscious visual continuity, despite minutes of blink-related blackouts each day[137–139]. Similarly, the frequency and timing of blinks are centrally modulated, which probably constitutes an adaptive behaviour to minimize loss of visual information during salient moments[140–144]. Blinking is an accessible physiological signal, easily observed with the naked eye, a camera system or skin electrodes placed near the eye.

Conscious level and state. Blinking predicts conscious level across multiple temporal scales and conscious state domains. In clinical

settings, spontaneous blinking differentiates wakeful unresponsive syndrome from the minimally conscious state[145,146]. Spontaneous blinking also correlates positively with clinical measures of conscious level[147], and blink rate over time (that is, days to weeks) predicts recovery of conscious awareness in disorders of consciousness[148]. In healthy individuals, blinking corresponds with attention and arousal and closely follows fluctuations in conscious level. For example, blink rate is altered by fatigue[149–152], sleep deprivation[153,154] and arousal (for example, emotional states of anxiety and anger, and drug-induced depressant states)[155]. Also, the timing of blinks corresponds with attentional breaks (for example, scene cuts during a film) that can serve to indicate transitions in attentiveness[156]. Furthermore, changes in blink rate are observed during altered states of consciousness, including hypnosis, mind wandering and 'blankness of mind'[155,157–160].

Conscious content. Spontaneous blinking can track the perceived content of bistable images and the occurrence of perceptual switches[161–164]. Although indication of a perceptual switch by blink rate alone cannot fully indicate conscious content, combined with knowledge of experimental stimuli and presentation design, perceptual content can be inferred. Blinking is also sensitive to various perceptual features, including image contrast and spatial frequency[165], pain[166], affective arousal (for example, blink magnitude modulated by viewing pleasant and unpleasant images)[167,168], time perception (for example, the perceived duration of visual and auditory stimuli)[169] and familiarity[170]. Furthermore, blink rate changes are linked to shifts between internally and externally oriented attention, providing clues about whether conscious content is focused on external sensory events or internal thoughts and emotions[171–173].

Finally, spontaneous blinking is modulated by sensory conscious perception. For example, blink rate increases after the conscious perception of near-perceptual-threshold visual and tactile stimuli, but it remains unchanged when the same stimuli are not perceived[3,83]. A similar finding was that spontaneous blinking predicted the moment of conscious perception of a target stimulus during continuous flash supression[174].

Eye movements

There are several major categories of eye movements, including smooth pursuit and saccades[175]. A primary function of eye movements is to stabilize the retinal image and focus light on the fovea. Fixational eye movements also prevent visual adaptation and improve visual acuity[176,177]. Artificially stabilizing the retinal image results in perceptual fading and blindness[178]. Numerous subcortical and cortical networks are involved with regulating eye movements[179,180]. This broad neural basis explains how eye movements are correlated with a wide range of behavioural, cognitive and perceptual processes[175]. Similar to the perceptual suppression that occurs during blinks, central neural regulation of eye movements partially functions to inhibit the conscious perception of visual blur during eye movements[181,182]. Furthermore, there may be shared neural substrates for eye movements and conscious perception, particularly in the frontal eye fields, which are linked with detection of visual stimuli and regulating saccades[65,183–185]. Eye movements are easily measured with a camera system and skin electrodes placed near the eye.

Conscious level and state

Eye movements are linked to conscious level and state. Clinically, people with impaired conscious level display distinct eye movement patterns, which are often used to differentiate among the disorders of consciousness and predict recovery[186,187]. For instance, eye movements in response to tactile stimulation[188,189], fixations (for example, more on-target fixations in minimally conscious state versus unresponsive wakefulness syndrome)[190] and visual pursuit (that is, tracking moving objects)[191,192] are key indicators of conscious level in disorders of consciousness.

In healthy individuals, altered states of consciousness are also predicted by eye movements. Eye rolling and changes in eye movement speed are common in hypnosis[160,193,194], whereas a decrease in microsaccade rate is associated with self-reported absorption[195]. Drug-induced altered states, such as those induced by hallucinogenic agents, are associated with increases in saccade rate and amplitude[196] (Fig. 3a).

Sleep stages are another consciousness-linked state domain for which eye movements provide insight. Rapid eye movement (REM) sleep — characterized by its flickering eye behaviour — is linked to vivid episodic dreaming[197]. Meanwhile, during non-REM sleep, the eyes typically rotate upward (during REM sleep the eyes rotate downward) and undergo slow, oblique drifts[150]. Eye movements are also altered in states of suppressed conscious level (for example, prolonged periods of fixation and decreases in saccade and microsaccade velocity), including sleep deprivation[198,199], fatigue[151,200,201] and drug-induced sedation[202]; these altered movements often correspond with subjective reports of fatigue and drowsiness.

Eye movements also reflect perceptual sensitivity and attentional states linked to conscious level. For example, saccade rate predicts the contrast detection threshold on a trial-by-trial basis[203]. Moreover, microsaccade rate and direction are modified by covert attention (for example, microsaccades are biased towards the location of task-relevant stimuli)[204,205]. In visual spatial neglect — an attentional deficit to a portion of the visual field — gaze position is biased towards the intact visual field, so eye movements can be used to infer the areas in visual space that are perceived and not perceived[206]. However, another study revealed that some individuals with visual spatial neglect exhibited a bias in saccade direction independent of visual perception[207].

Conscious content

Fixational eye movements are relevant for predicting and potentially regulating (for examples, see refs. 30,163) conscious content. For instance, they are linked to perceived perceptual motion (real and illusory)[161,208–210], image orientation (for example, saccade rate increases at the time of perceptual switches)[162] and heading perception (that is, direction of self-motion)[211]. The fixation-saccade pattern or scan path also corresponds with image and scene content[212]. Eye movements even predict conscious content during dreams — reflecting the dreamed viewing field and direction (for example, a downwards eye movement linked with the dream content of picking up an object from the floor)[150,213–216]. Similarly, eye movements during imagery reflect the same fixations and scan paths as when viewing a physical image[217–220]. However, eye movements during imagery were found not to predict perceptual vividness[217].

In addition, eye movements are sensitive to low-level perceptual features (for example, image contrast and spatial frequency)[165,221], duration perception[222,223], image position (for example, images appear mislocalized and scene images appear compressed when presented just before a saccade)[224], emotions (for example, microsaccade rate is lower for self-reported emotionally unpleasant than pleasant images)[122] and perceived motion direction[23,78,225].

Finally, eye movements are sensitive to sensory conscious perception. Microsaccade rate is suppressed — a phenomenon known

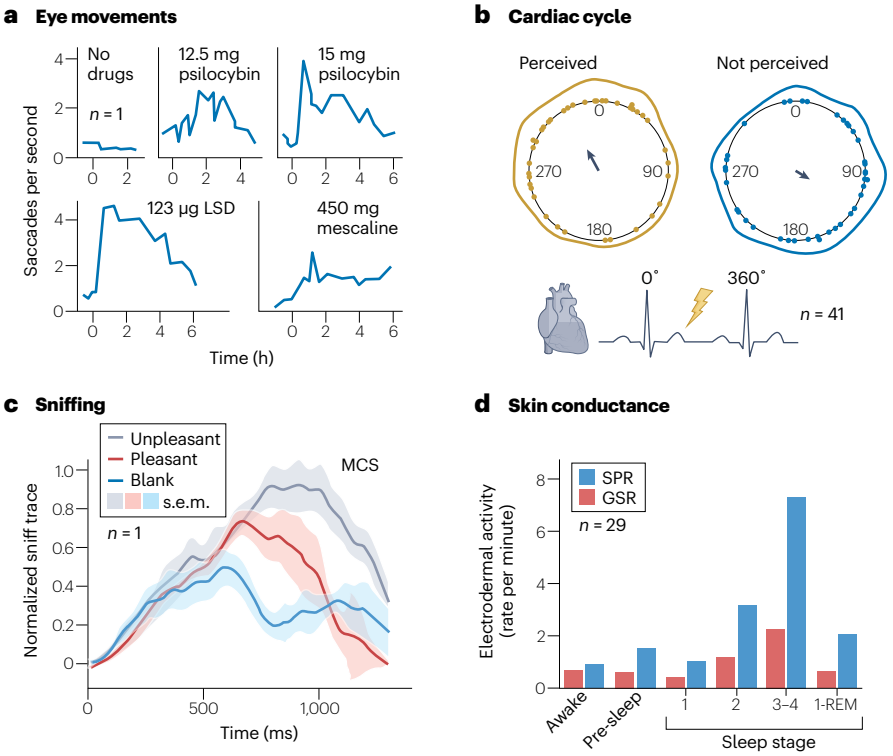


Fig. 3 | Physiological signals used to infer conscious level and state. **a**, Saccade rate change over hours for one participant after ingesting 12.5 and 15.0 mg of psilocybin, 123 µg of lysergic acid diethylamide (LSD) and 450 mg of mescaline[196]. A no-drug condition is also shown. These results highlight that there is a relationship between eye movements and psychedelic states. **b**, Perceived versus not perceived finger stimulation relative to the cardiac cycle, revealing that cardiac phase corresponds with perceptual sensitivity. Each dot represents one participant ($n = 41$)[333]. The centre arrow direction and length indicate the group average mean angle and mean resultant length R from a Rayleigh test of uniformity, respectively, for perceived (hit) and not perceived (miss) events. The Rayleigh test of uniformity was statistically significant for perceived events ($R = 0.34$; $P = 0.007$). However, another study did not find that cardiac phase influences conscious perception, but it did report that cardiac phase affects brain activity linked to conscious processing[331]. **c**, Mean and standard error of the mean (s.e.m.) traces of sniffing volume for unpleasant, pleasant and blank odour conditions in a patient in a minimally conscious state (MCS)[364]. Odour stimulation was presented at 0 ms. Sniffing in an patient with unresponsive wakefulness syndrome (data not shown) did not discriminate among the odour conditions. This result supports the idea that conscious state (MCS versus unresponsive wakefulness syndrome) can be inferred from sniffing behaviour. **d**, Spontaneous skin potential responses (SPRs) and galvanic skin responses (GSRs) during wakefulness, pre-sleep and different sleep stages (stages 1, 2 and 3–4 and 1-rapid eye movement (REM) sleep) ($n = 29$)[277]. This finding suggests that skin conductance can be used to infer sleep-linked conscious level. Part **a** adapted from ref. 196, Springer Nature Ltd. Part **b** adapted with permission from ref. 333, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **c** adapted from ref. 364, Springer Nature Ltd. Part **d** adapted with permission from ref. 277, Wiley.

as oculomotor freezing or microsaccade inhibition — following the conscious perception of near-perceptual-threshold visual or tactile stimuli[3,83,226,227]. Similarly, increases and decreases in microsaccade rate and magnitude predict perceptual appearances and disappearances, respectively[29]. Oculomotor freezing occurs during illusory perception (that is, false alarms or reporting a stimulus when none appeared)[228]. Additionally, fixational eye movements are reduced by the temporal expectation of predicable sensory events, including in vision, audition and tactile conscious perception[144,229].

Muscle signals

Overt report relies on muscle activity, including volitional speech and body movement (Fig. 1a). The absence of muscle activity (for example, in cases of paralysis or cognitive–motor dissociation) is a major impediment for detecting consciousness, because key behaviours used for overt report, such as speech and body gestures, are lost[230]. Still, covert muscle activity may be present without visible behaviour. Although imperceptible to the naked eye, subtle muscle signals are detectable with camera systems and dermal sensors that record the electrical discharges associated with muscle activation. In addition, muscle activity metrics, such as reaction time and power, are difficult to precisely measure visually. Thus, even overt muscle activity contains covert signal features. Muscle signals are largely regulated by central neural processes, making them useful proxies for neurophysiological states and events that are relevant in behaviour, cognition and perception.

Conscious level and state. Careful observation (for example, reviewing slowed-down video footage) can reveal subtle movements, such as microexpressions, that may indicate conscious level and state[12,231]. For instance, in people unable to self-report, facial and body movements predicted the presence of subjective pain[232]. Similarly, non-verbal behavioural responses to noxious stimulation (for example, grimacing,

brow lowering and non-reflexive defensive behaviours) distinguish different disorders of consciousness[233–237]. Non-pain-related behaviours, including leg crossing[238] and swallowing[44,239], also discriminate conscious level in such disorders. Nose scratching and sheet grabbing may, too, be relevant to infer conscious level in patients with impaired consciousness[240]. Additionally, covert muscle signals measured with electromyography (EMG) could be used to infer conscious level and state, including the detection of intraoperative awareness[241].

Conscious content. Emotion conscious content can be inferred by overt muscle activity, particularly facial expressions (for example, smiling and frowning)[242]. Subtle or covert facial muscle signals captured by still images, videos or facial EMG can also be used to infer conscious emotion, including anxiety, fear, pleasure and displeasure, disgust and joy[243–246]. Facial EMG responses can also distinguish between pleasant and unpleasant arousal[167]. Even imagery of affective positive and negative scenarios evokes facial EMG responses such as those associated with overt displays of pleasure, sadness, fear and anger[247–249]. Additionally, facial EMG has distinguished between people with and without depression while they viewed affective imagery, with the former showing more sad-like EMG responses[250].

Covert muscle signals linked to the production of overt vocalization support inferences about conscious content. In one study, participants were prompted to say a phrase during lucid dreaming that produced a unique EMG signature when spoken aloud[251]. These EMG signatures were detectable during dreaming, corresponding with the participant's report of successfully speaking the phrase in their dream. Similarly, lucid dreamers were able to perform pre-instructed mouth movements while dreaming[252]. These findings build on previous studies revealing that facial EMG signatures can distinguish between conversational and non-conversational dreams[253,254]. In addition, EMG signals from limbs are reported to correspond with dreamed limb movement[255], and small body movements during REM sleep have been found to predict dream content[256,257].

A valuable metric of muscle activity is reaction time, because it is linked to behavioural, cognitive and perceptual processes. For instance, reaction time predicted imagined rotation during a shape-matching task[258]. In particular, the timing of conscious versus unconscious processing may be distinguishable and thus represented in motor behaviours (that is, early and late neural dynamics correspond with unconscious and conscious processing, respectively)[259]. For example, using the redundant signal effect (that is, quicker stimulus-evoked responses for multiple stimuli versus a single stimulus), reaction times are quicker according to conscious perception of the duplicate stimuli[260]. A study with humans and non-human primates showed that reaction times during a spatial-cueing task discriminated between perceived and not perceived cues[261]. Indication of conscious perception via reaction time and knowledge of the cue stimulus makes it possible to infer conscious content.

A nascent area of research involves auricular muscles of the human ear. In some non-human animals, ear movements are overt (for example, a deer pointing its ears in the direction of a sudden sound). In humans, ear movements are largely absent[262]. Yet, one study found that muscle activity in the vestigial human auriculomotor system predicted the perceived direction of sound[263] (Fig. 4b).

Dermal signals

Despite the static appearance of the skin, skin or dermal physiology is dynamic, including changes in temperature and conductance related to sweating. Whereas eye movements and muscle signals an be recorded from the skin, this review considers dermal signals as a separate category of physiology. Dermal signals are valuable because they are easy to record, cannot be voluntarily controlled[264], are sensitive to multimodal sensory stimulation and are linked to neurophysiological activity. For instance, skin conductance has been used to monitor the autonomic nervous system[265]. As a result, dermal signals offer insight into central neural activity, allowing inferences on processes linked to conscious level, state and content.

Conscious level and state. Skin conductance or electrodermal activity is a potential proxy for conscious level owing to its sensitivity to enabling factors of consciousness, including arousal and attention[266]. In support of this, skin conductance dynamics (for example, in response to visual stimulation) can predict the conscious level clinical scores of patients[267,268]. One study even found that electrodermal activity evoked by emotional words distinguished conscious levels in people with brain injuries[269]. However, some studies reported that skin conductance does not distinguish conscious levels in disorders of consciousness[270,271]. In addition, skin conductance fluctuations can be used to monitor conscious level under general anaesthesia and to determine intraoperative awareness[272,273]. Notably, skin conductance was found to be more sensitive than clinical stress scores (that is, markers of intraoperative physiological stress, including blood pressure >130 mmHg and cough) for determining conscious level[274].

Normal variations in conscious level and state (for example, sleep) are also reflected in dermal signals, including distinguishing REM (skin conductance inhibited) and non-REM sleep (skin conductance excited[275–278]; Fig. 3d). Altered states of consciousness, such as hypnosis and mediation, are characterized by decreases in skin conductance relative to baseline[279]. In addition, people in drug-induced altered states (for example, psychedelic-like conscious content induced by nitrous oxide inhalation) show changes in skin conductance responsiveness to auditory stimulation that are associated with subjective reports (for example, stimulus pleasantness)[280,281].

Conscious content. Dermal signals are particularly sensitive to interoceptive conscious content. For example, skin conductance corresponds with nociception (unconscious pain) and conscious pain[114,282–284]. Skin conductance can also be used to infer the occurrence of specific emotions (for example, those experienced when imagining or viewing affective images)[167,168,285,286], including the emotional content of dreams[287]. Skin conductance is further linked with satiety (note that skin temperature is also linked to satiety)[288] and interoceptive stimulation (for example, the perceived intensity of oesophageal stimulation)[289].

Dermal signals are also informative of the conscious content related to self-consciousness, including perceived body and thought ownership (that is, embodiment and agency). Self-consciousness is impaired in various psychiatric and neurological disorders[290]. In addition, people with amputated limbs can experience altered states of body ownership[291]. In healthy individuals, altered states of self-consciousness can be experimentally induced using procedures (for example, the rubber hand and enfacement illusion) that create the illusion of ownership of a foreign object (for example, a table or artificial hand) or body part. Skin conductance responses predict the vividness of these illusions and differentiate between painful and non-painful touch applied to the illusory incorporated object or body part[292–294]. Similarly, skin temperature changes, such as a drop in temperature for the limb targeted in a body ownership illusion, have

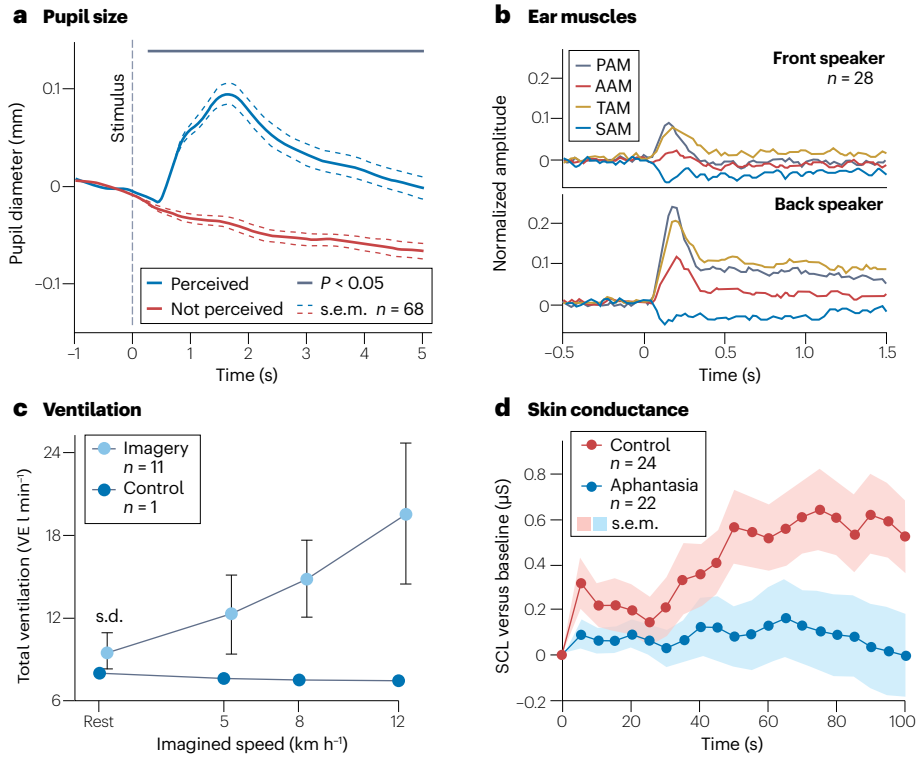


Fig. 4 | Physiological signals to infer conscious content. a, Group mean ($n = 68$) and standard error of the mean (s.e.m.) pupil size change for perceived and not perceived visual near-perceptual-threshold stimuli[3]. Stimuli were presented at 0 s. This result highlights how pupil size can be used to infer conscious perception of visual stimuli and has been replicated in non-visual (for example, tactile[83]) conscious perception. Conscious content is inferred from conscious perception on the basis of prior knowledge of the stimulus type (a face) and presentation time (0 s). **b**, Group mean ($n = 28$) normalized ear muscle responses to ipsilateral front and back speaker (relative to the participant) auditory stimulation[263]. This finding supports the idea that the perceived location of auditory stimulation is linked to ear muscle activity. **c**, Group mean ($n = 11$) and standard deviation (s.d.) of total ventilation (pulmonary ventilation (VE)) for imagined rest and locomotion at different speeds (5, 8 and 12 km h^{-1}) versus a non-imagery control ($n = 1$)[345]. Accordingly, the conscious content of imagined locomotion corresponds with ventilation signals. **d**, Group mean and s.e.m. skin conductance level (SCL) change for imagery of frightening stories in control participants ($n = 24$) and people with aphantasia (people with low or no imagery ability; $n = 22$)[307]. This result reveals that vivid versus dull imagery conscious content is distinguishable by skin conductance. AAM, anterior auricular muscle; PAM, posterior auricular muscle; SAM, superior auricular muscle; TAM, transverse auricular muscle. Part **a** adapted from ref. 3, Springer Nature Ltd. Part **b** adapted from ref. 263, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **c** adapted with permission from ref. 345, Elsevier. Part **d** adapted with permission from ref. 307, Royal Society.

been linked to the vividness of the illusion[295–297]. This suggests that dermal signals may be linked with the neural processes underlying the conscious perception of body ownership[298].

Dermal signals can also predict sensory conscious perception. For example, skin conductance magnitude is lower for not perceived relative to perceived near-perceptual-threshold images[299]. Moreover, conscious knowledge of stimulus presentation rules, such as the occurrence of an aversive sound predicted by a preceding sound sequence, can also be inferred from changes in skin conductance magnitude[300].

Finally, dermal signals predict various sensory-specific conscious content features, including distinguishing among sound features (for example, skin conductance magnitude changes relative to timbre and rhythm)[301], image content (real and imagined; for example, emotion or valence, saturation, brightness and colour)[302–305], tactile stimulation intensity (for example, larger and prolonged changes in skin conductance for high-intensity touch)[115] and olfactory stimulation (for example, distinguishing smells that do and do not induce craving)[306].

Skin conductance has also been linked to mental imagery vividness, with one study showing that people with normal imagery abilities exhibit distinct skin conductance changes during arousing mental imagery, whereas individuals with aphantasia show little-to-no skin conductance change[307] (Fig. 4d).

Cardiovascular signals

The cardiovascular system transports oxygen, nutrients and waste products throughout the body via blood. The heart is the essential pump that drives blood through the vascular system. Whereas cardiac conduction operates autonomously, cardiac activity is modified by the CNS[308]. Conversely, brain activity is influenced by the cardiac cycle, for example, influencing cortical excitability, cerebrospinal fluid flow and the direct pulsating of brain tissue[309]. These interactions between central neurophysiology and the cardiovascular system explain how cardiovascular function is both informative of and influenced by behaviour, cognition and perception. Cardiovascular signals, including heart rate, heart rate variability and cardiac cycle, are readily accessible using

electrocardiography and photoplethysmography, which is now a common feature in many wearable devices.

Conscious level and state. The cardiovascular system is influenced by cortical and subcortical regions, including the insula, thalamus and medulla — all areas implicated in neural arousal and conscious level[3,308]. When normal arousal state is impaired by traumatic brain injury, cardiac signals change, suggesting that cardiovascular activity can serve as a proxy for central neural states relevant to establishing conscious level[310]. For instance, heart rate and heart rate variability distinguish between the minimally conscious state and unresponsive wakefulness syndrome (for example, there is a positive correlation between heart rate and conscious level clinical scores in unresponsive wakefulness syndrome)[311,312].

Furthermore, general anaesthesia affects cardiac activity, including by decreasing blood pressure, heart rate variability and heart rate[313,314]. Likewise, cardiovascular signals (for example, heart rate and blood pressure) are relevant for monitoring the depth of anaesthesia and intraoperative awareness[315]. Altered states of consciousness, including drug-induced hallucinatory states, hypnosis, trance and meditation correspond with unique cardiovascular signals, including changes in blood pressure, heart rate, heart rate variability, and cerebral blood flow[73,279,316–320].

During sleep, cardiac dynamics distinguish between REM and non-REM sleep, with increased heart rate during REM sleep[321–325]. In addition, a study found that heart rate variability and blood pressure distinguished between people who remembered their dreams upon waking and those who did not[326]. The occurrence of lucid dreaming is also marked by increases in heart rate relative to non-lucid dreaming REM sleep[327].

Sensory-evoked neural activity is altered by the cardiac cycle and activity (for example, heart rate and cardiac phase), implying a possible relationship between the cardiac system and the sensitivity state of sensory conscious perception[328–331]. For example, near-perceptual-threshold stimuli are perceived more often during diastole and less often during systole[328,332,333] (Fig. 3b), although there is at least one study that did not find this relationship[331]. Also, lower heart rate or heart rate deceleration correlates with enhanced perceptual performance[334,335]. Furthermore, when visual stimuli were presented synchronously to the cardiac frequency, they took longer to break from perceptual suppression (that is, continuous flash suppression)[336]. Similarly, lower pain perception is reported when noxious stimulation is presented during systole versus diastole cardiac phases[337]. This result is possibly explained by certain phases of the cardiac cycle interfering with sensory processing, transiently inhibiting conscious perception[309]. In support of this interpretation, a masking stimulus presented during systole allows more frequent detection of a target stimulus than occurs when the same stimulus is presented during diastole[338]. This finding can be explained by conscious suppression of target stimuli by the mask being less effective during systole.

Conscious content. Cardiovascular signals are sensitive to interoceptive conscious content. For example, pain perception is linked to changes in heart rate and blood pressure[284,339,340]. Importantly, when noxious stimulation is not perceived (for example, in cases of spinal cord injury), these cardiac signals are absent[341]. Cardiovascular signals are also sensitive to affective arousal. For instance, the viewing of arousing images (for example, erotic or homicide images) corresponds with a decrease in heart rate and vasoconstriction[168]. Likewise, a study showed that heart rate changes differentiated between viewing pleasant and unpleasant arousing images[167]. Also, engagement of affective imagery results in heart rate changes (for example, higher heart rate for high-arousal than low-arousal imagery)[249,305,342]. Similarly, imagined and dreamed exercise modulated heart activity in line with how physical exertion affects cardiac signals[343–345]. Likewise, dream intensity and affective conscious content were linked with heart rate and heart rate variability[287].

Furthermore, heart rate decreases were found after perceived versus not perceived stimuli, so it is possible to predict conscious content if the type and timing of stimulus presentation are known[346]. Cardiac signals an also be used to infer various domains of conscious content, including colour (for example, heart rate deceleration and acceleration for achromatic and chromatic stimuli, respectively)[303], sound timbre, intensity and duration[301], and time perception (for example, lower heart rate correlated with judging an auditory stimulus as longer)[347,348]. However, in the specific instance of time perception, cardiac activity was found to be predictive only through its link with arousal (that is, heart rate modulation alone was not sufficient to affect time perception)[349].

Respiratory signals

Respiratory and cardiovascular activity are highly interdependent[350]. Prominently, the respiratory sinus arrhythmia is the heart rate beat-to-beat variability linked to the respiratory cycle[351,352]. Like cardiovascular function, the respiratory system is mutually influenced by central neurophysiology[353–356]. These respiratory–brain interactions explain numerous connections between respiration and behaviour and cognition[357,358]. In addition, respiratory dynamics influence perception, particularly odour perception[359,360]. Respiratory signals are easily recordable using plethysmography (that is, a belt sensor that measures changes in chest volume), nasal probes (for example, recording temperature or airflow as a proxy for breathing cycle) and camera systems (for example, recording chest movements or thermal changes near the nose and mouth)[361,362].

Conscious level and state. Respiratory dynamics are partially regulated by the arousal network, including brain regions closely linked to arousal and conscious perception[3]. Several studies have used respiratory activity as an indicator of conscious level in disorders of consciousness[363]. For example, sniff volume for pleasant and unpleasant odours — which is typically inhibited for unpleasant odours — distinguished between patients in a minimally conscious state and those with unresponsive wakefulness and predicted recovery of consciousness[364] (Fig. 3c). This result also highlights how conscious content of perceived odour pleasantness can be inferred from sniff volume and duration (see below). Sniffing has also been used to assess the ability to follow commands and communicate in disorders of consciousness (for example, sniff volume change in response to a cue), suggesting a potential method of overt communication similar to mental imagery-based approaches[365,366].

Respiratory signals can also distinguish between wakefulness and sleep, and among sleep stages, with individuals exhibiting shallower, faster and more irregular breathing during REM sleep than non-REM sleep[367,368]. Likewise, drowsiness can be predicted by a change in respiratory activity (for example, a decrease in mean breathing rate)[362]. Also, a study showed that the transition from alpha to theta band and theta to alpha band electroencephalography activity, indicating shifts between sleep and wakefulness, corresponded with changes in breathing depth,

with shallower breathing versus deeper breathing in transitions to sleep and wakefulness, respectively[369].

General anaesthesia also induces respiratory changes, which vary by drug type and correspond to the depth of unconsciousness[370]. Moreover, an increase in respiration rate has been recorded during drug-induced hallucinogenic conscious states (such as that induced by psilocybin) relative to placebo[73]. Rest and trance conscious states can be discriminated by respiratory activity[317]. Likewise, hypnotic and meditative altered states of consciousness correspond with a decrease in respiration rate[279].

Respiratory signals are also linked to sensory perception. For example, a study found that visual stimuli were more frequently detected during exhalations[371]. Another study showed that respiration phase influences perceptual sensitivity for near-threshold visual stimuli[372]. This effect may be related to respiratory-linked changes to neural excitability and exteroceptive–interoceptive attentional shifts — exteroceptive attention being dominant during inhalation and interoceptive attention being dominant during exhalation[358,373,374]. Relatedly, the pain perceived following an electric shock was reduced when the shock was delivered during inhalation rather than exhalation[375].

Conscious content. Respiration signals are sensitive to conscious content related to self-consciousness and interoception. For instance, respiration behaviours can indicate conscious emotion[376]. Likewise, overt expressions of respiratory activity, such as gasps and yawning, are often interpreted as signs of emotional state (for example, fear or boredom). More subtle signals — that is, breathing rate and depth — correspond with negative versus positive emotional reaction to different images and sounds[377,378]. Even the perception of time (for example, estimating for how long an image was presented on a screen) is influenced by the preceding breathing dynamic, with image duration being estimated as shorter following paced versus normal breathing[349,379].

Respiratory activity is sensitive to conscious content related to odours. Sniffing is closely linked with olfactory perception[380], and specific respiratory patterns correspond to the conscious perception of odours[381,382]. When an odourant was administered intravenously, an odour was perceived only when the participant made a sniff or air was blown into their nose[383]. Sniffing has also been associated with phantosmia or olfactory hallucinations in the absence of an odourant[384]. Correspondingly, sniffing for perceived pleasant versus unpleasant odours is distinguished by respiratory frequency and sniffing vigour and volume[364,381,385].

Even in the absence of odourants, odour-related imagery induces sniffing behaviours. Imagined odour content corresponds to inhalation volume, with greater inhalation volumes associated with pleasant than unpleasant imagined odours[386]. Sniffing behaviour during the viewing of odour imagery also discriminated participants with dull versus vivid odour imagery ability[387]. Additionally, non-odour-related imagery of different levels of exercise vigour (for example, imagined walking at different speeds) was distinguishable by respiratory activity[345] (Fig. 4c).

Dream content has also been linked to respiratory activity. Higher respiratory rates have been found to predict dreaming upon awakening from REM sleep[216]. In studies of lucid dreamers, participants were able to enact respiratory behaviours during their dream (for example, speaking, yodelling or physical exercise) that closely matched the awake, overt engagement of these respiratory-linked actions[255,344,388]. Together, these findings provide evidence that respiratory signals can track conscious content across various states of consciousness, independent of sensory stimulation.

Perspectives

Although covert measures of consciousness are typically discussed in the absence of overt report, such as in cases of paralysis or no-report studies, combining covert measures with overt report may enable more reliable and detailed inferences of subjective experience and state, potentially addressing some limitations of overt report (Fig. 1b). For example, patient pain, which is often measured with self-reported pain scales[389], could be corroborated with covert measures of pain (for example, facial EMG or electrodermal signals). Additionally, covertly detected conscious level, state and content during resting-state studies could be compared with retrospective reports by participants on their conscious experiences during rest.

Studying the coherence between covert measures and overt report can be valuable. For instance, pupillary responses have been found to differentiate between people with and without imagery ability[91]. However, in the same study, pupil size did not correlate with self-reported imagery vividness. Similarly, whereas pupil size predicts objective markers of fatigue (for example, reaction time), it correlates with subjective sleepiness only at certain stages of the circadian rhythm[69,390]. Likewise, gaze position has been shown to be more sensitive than standard self-reported neuropsychological tests for measuring visual neglect after right-hemispheric stroke[206].

Another promising area of research involves exploring human covert measures of consciousness in non-human animals. Non-human animal consciousness remains a controversial and active area of study[7]. Applying human covert measures of consciousness to non-human animals may provide insights into their capacity for subjective states. Some physiological measures are applicable across species (for example, blink rate corresponds with arousal state in crows and horses[391,392]). However, other physiological signals undergo species-specific dynamics (for example, with an increase in arousal, pupils constrict in pigeons but dilate in primates and rodents[393]) or may be absent altogether in some species (for example, electrodermal signals in species without sweat glands). Therefore, species-specific physiological measures may be necessary to address the heterogeneity of physiological traits across animals. For instance, skin colour and texture are relevant physiological signals of arousal and sleep stage in octopuses[394,395].

The challenges and limits of covert measures of consciousness

A major challenge for discovering covert measures of consciousness is distinguishing physiological signals linked to conscious versus unconscious neural processes (Fig. 5). Although the highlighted physiological signals show evidence that they are associated with conscious level, state and content, they may instead reflect neural processes that occur without consciousness. For instance, changes in pupil size (which can differ, for example, when an individual is presented with happy versus fearful images) are recorded in the absence of conscious perception[118,396,397]. Moreover, the pupillary light reflex is preserved in the absence of conscious vision[398,399]. In addition, changes in pupil size and blood pressure under general anaesthesia are not necessarily indicative of awareness[400], and sniffing and skin conductance dynamics can occur without conscious awareness[302,401–405]. Although reddening of the face or flushing may signal conscious pain, it is also observed in patients suspected to be unconscious[235]. Affective images shown in the blind field of people with cortical blindness evoke pupillary and facial EMG signals that are similar to those evoked by images shown in the sighted field[406]. Conversely, conscious awareness may persist even when physiological dynamics suggest unconsciousness (for example, inhibited breathing under general anaesthesia)[188,370,407]. Thus, it remains

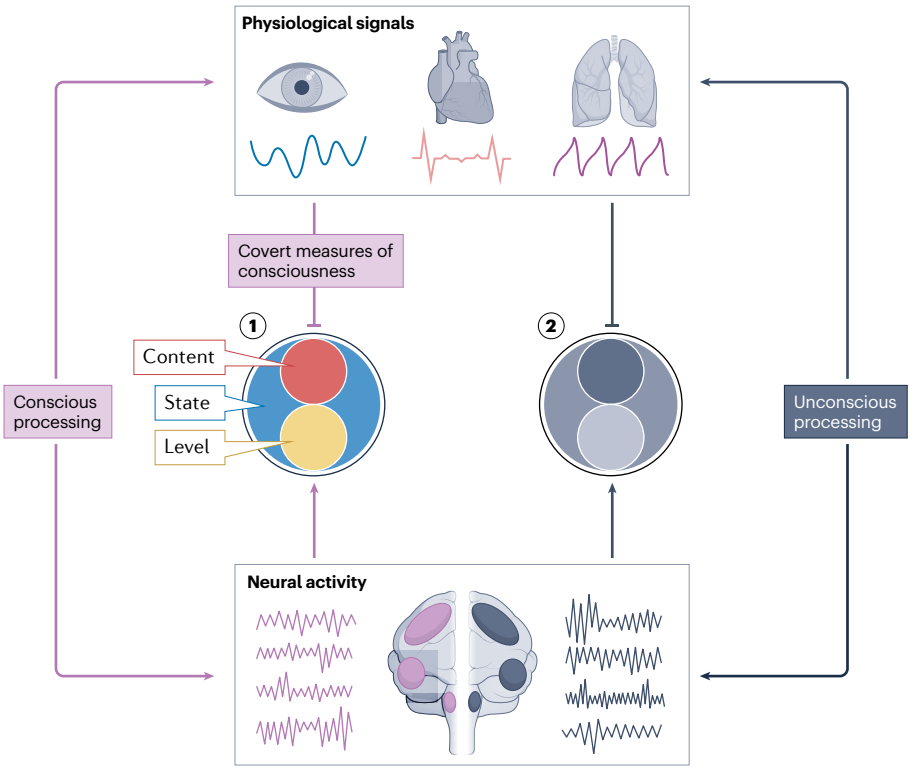


Fig. 5 | Distinguishing physiological signals indicative of conscious and unconscious processing. Neural activity and physiological signals, such as pupil size, cardiac responses and respiratory activity, are mutually influenced (indicated by the bidirectional arrows). Consciousness may or may not be linked to such neural activity. Here, conscious processing refers to neural activity that is indirectly (for example, neural activity underlying enabling factors, including neural states of arousal and attention) or directly linked to the neural mechanisms of consciousness, whereas unconscious processing refers to neural activity that does not influence consciousness or occurs in the absence of consciousness. A covert measure of consciousness is a physiological dynamic that offers insight into subjective experience, particularly among the dimensions of conscious level, state and content, as it is either influenced by or exerts influence on conscious processing. A major challenge in identifying covert measures is differentiating physiological signals that indicate (1) when consciousness and its subdomains — level, state and content — are present versus (2) when consciousness is absent. Notably, physiological signals can reflect both conscious and unconscious processes (for example, markers of arousal can be present with and without consciousness), making it challenging to determine the specificity of covert measures for consciousness. A similar challenge applies to overt report when behaviour is observed without consciousness (Fig. 1b). This limitation must be carefully considered and tested when validating potential physiological markers of consciousness, for example, observing physiological signals under both conscious and unconscious conditions. Note that the highlighted brain nodes for conscious processing are only example regions and do not claim to represent the true locations of the brain networks necessary for consciousness.

unclear how to definitively discriminate physiological signals that indicate conscious experience from those that reflect unconscious processing or reflexes[187].

These examples challenge the specificity of candidate covert measures of consciousness. Likewise, there are numerous influences on physiological signals (for example, pupil size is linked with arousal, locomotion, salience, learning, decision-making and cognitive load[408]). Thus, our ability to know when pupillary dynamics or any other candidate covert measures are responsive to consciousness versus other sources of modification is currently limited. In this Review, we have been intentionally agnostic to the exact mechanism by which a covert measure is linked to consciousness. Indeed, there are likely to be variable and complicated linking mechanisms between covert measures and consciousness. Moreover, although there may be direct relationships between covert measures and consciousness (that is, the neural mechanisms of consciousness directly modify a physiological signal), there may also be indirect relationships (for example, a physiological signal is responsive to enabling factors or the consequences of consciousness). Whether through a direct or an indirect process, a signal can be designated a covert measure. Nonetheless, elucidating the precise mechanisms that link covert measures with consciousness is valuable to better understand what details covert measures can supply about consciousness and its physiological precursors, sources and consequences.

One approach to assess the specificity of physiological signals for consciousness is to test these measures in unconscious or subliminal conditions. For instance, perceptual-threshold paradigms involve presenting identical physical stimuli across two perceptual conditions: (1) perceived and (2) not perceived. Demonstrating that a physiological signal is uniquely responsive to the perceived condition provides evidence that it represents conscious processing (for examples, see refs. 3,83). In short, whenever feasible, conscious and unconscious conditions must be tested for all candidate physiological signals to confirm each signal's unique association with consciousness, independent of

Glossary

Conscious content
The subject or category of subjective experience; also called qualia, phenomenology or phenomenal experience.

Conscious level
The likelihood of being or becoming conscious, closely linked with the physiological states of arousal, alertness and vigilance.

Conscious state
The unique combination of conscious level and content characteristic of specific states of consciousness.

Covert measures of consciousness
Physiological signals linked to conscious level, state or content (for example, pupil size, eye movements and skin conductance).

Enabling factor of consciousness
Prerequisite or precursor physiological states (for example, arousal and attention) that facilitate consciousness. These factors are often necessary but not sufficient for consciousness.

unconscious processing. Nonetheless, in some experimental settings, it is possible to use nonspecific physiological measures to infer conscious level, state and content. For example, the pupillary light reflex, although ambiguous for conscious vision, can predict the perceived image during binocular rivalry of high versus low luminance images[78]. Similarly, the environmental and behavioural controls available in an experimental context can limit the various consciousness-independent factors that also drive physiological dynamics.

However, covert measures are limited because of their frequent reliance on prior knowledge of the sensory environment and participant behaviour to interpret physiological signals. For example, information on experimental parameters, including stimulus features (for example, task relevance, valence, frequency, onset and duration) and task behavioural requirements (for example, participant instructions and reporting procedures) are essential for inferring conscious content from physiological signals that indicate conscious access (for example, perceived versus not perceived). Therefore, a key future ambition is to develop covert measures of consciousness that require minimal prior knowledge of sensory parameters and task demands. This advancement would expand the applicability of covert measures of consciousness, including enabling their use in less controlled or entirely unknown sensory and behavioural contexts.

Finally, experimenters and clinicians should be aware that covert measures of consciousness can be altered in patient populations, thus impacting the reliability of these measures across groups of people (for examples, see refs. 207,227). There is even individual variability in the responsiveness of covert measures. For example, a study of visual stimulus-evoked pupil size, blink and microsaccade dynamics found individual differences in the reactivity of these eye metrics for perceived images[227]. For this reason, the physiological signals of interest should be tested within the target population and multiple covert measures should be assessed within individuals.

Future directions

Covert measures of consciousness are a nascent development. There are numerous future directions for evaluating covert measures and demonstrating their possible applications, in part, urged by the limitations highlighted above. The top-priority future aims include (1) isolating the features of covert measure physiological signals linked to consciousness versus other enabling factors and consciousness-independent influences (for example, conscious versus unconscious processing; Fig. 5); (2) examining the combination of multiple physiological signals to leverage orthogonal predictive value (for example, combining eye metrics improves the prediction of perceived stimuli compared to relying on any one metric alone[3,227]); (3) testing the generalizability of covert measures at the individual level (for example, variable physiological responsiveness within individuals), group level (for example, variable physiological responsiveness between healthy and clinical populations) and environmental level (for example, variable physiological responsiveness according to task design and experimental context); (4) integrating covert measures of consciousness with overt report to determine best practices for complementary use to infer subjective states, including studying disagreements between overt report and covert measures that may help to elucidate the strengths and weaknesses of these approaches for inferring conscious level, state and content; (5) studying the brain activity linked to overt report and covert measures to reveal the extent of shared neurophysiological sources; and (6) determining whether the direct modification of covert measures (for example, with biofeedback[409] or behavioural interventions[349]) can causally influence conscious level, state or content.

Challenging the privacy of consciousness

Advances in neural recording and decoding — colloquially referred to as 'brain reading' — have introduced exciting new applications, including enabling people with paralysis to regain their ability to communicate and perform voluntary actions[28,49]. However, these technologies also raise ethical challenges, because they threaten the innate privacy of consciousness[410–412]. This potential intrusion has prompted discussion on how to safeguard the collection and use of neural and other physiological data[413,414].

Tempering the pace for developing brain reading methods are the technical limits of recording central neural activity. Many probes of neural activity require specialized equipment and facilities (for example, magnetoencephalography, MRI and positron-emission tomography), and some involve invasive procedures (for example, intracranial electroencephalography). By contrast, non-neural covert measures of consciousness are easier to record across diverse populations and environments. In fact, numerous commercial products and public technologies, such as smartwatches, virtual reality headsets, vehicles and video surveillance systems, already record many of the physiologies highlighted in this Review, including eye metrics, facial expressions, cardiovascular activity and dermal signals. Today, these physiological signals are analysed to provide feedback on health-related metrics (for example, resting heart rate, arrhythmia, maximal oxygen consumption and sleep behaviour). For example, a study of thousands of individuals analysed data from wearable devices to assess sleep patterns[415]. In the future, the same physiological data could be leveraged to infer consciousness. Furthermore, with modern computational approaches offering near-instantaneous analysis of physiological measures (for example, see ref. 416), it is increasingly possible for this information to be used in real time in scientific, clinical and commercial settings for the manipulation of attention, behaviour and consciousness. This application would not only be a challenge to the privacy of consciousness, but also undermine behavioural and phenomenal autonomy.

The ability to covertly access consciousness introduces new ethical concerns about the privacy of first-person experience. For instance, under what circumstances is breaching this subjective boundary

acceptable? Should car manufacturers be allowed to record physiological signals to alert drivers of drowsiness? In the event of an accident, should insurance companies have access to these data to guide determination of culpability? The judicial system may also be tempted to adopt covert measures as a new category of evidence in court[417]. Likewise, should these measures be used in a criminal investigation to infer relevant subjective states, such as to predict conscious level at the time of a crime or the familiarity of a suspect with key people or objects? Additionally, how should the government and law enforcement entities be expected to protect and analyse physiological data as they become more accessible by ubiquitous public video surveillance[418]? Reaching consensus on these and related questions is crucial and urgent as devices capable of recording physiology relevant to consciousness become widespread and modern analytical approaches, particularly machine learning, make their interpretation increasingly accessible and predictive of subjective states.

The potential advantages and harms of covert measures of consciousness must be carefully weighed. Although the subjective nature of consciousness is the major impediment for its detection and study, we must be cautious in dismantling this barrier that shields the first-person perspective from external observation. In pursuit of covert access to consciousness, we risk compromising the inherent privacy of subjective experience granted to all conscious beings.

Conclusions

Consciousness is private. The innate subjective nature of conscious experience presents numerous philosophical, scientific, clinical and ethical challenges. Overt report is the primary method by which we communicate our conscious experiences and infer the conscious experiences of others. However, overt report is limited because it may not always be accurate, consistently available or aligned with research goals. These limitations encourage the search for covert measures of consciousness: involuntary and accessible physiological signals that can provide insight into conscious level, state or content. Potential human physiological indicators of consciousness include eye metrics, skin signals and cardiovascular activity. Covert measures of consciousness can be used independently or in conjunction with overt report to achieve scientific and clinical goals. Future study of species-specific covert measures of consciousness may support the detection of consciousness in non-human animals. Finally, the advantages of discovering and applying covert measures must be balanced against the ethical implications of infringing on the privacy of the first-person perspective.

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